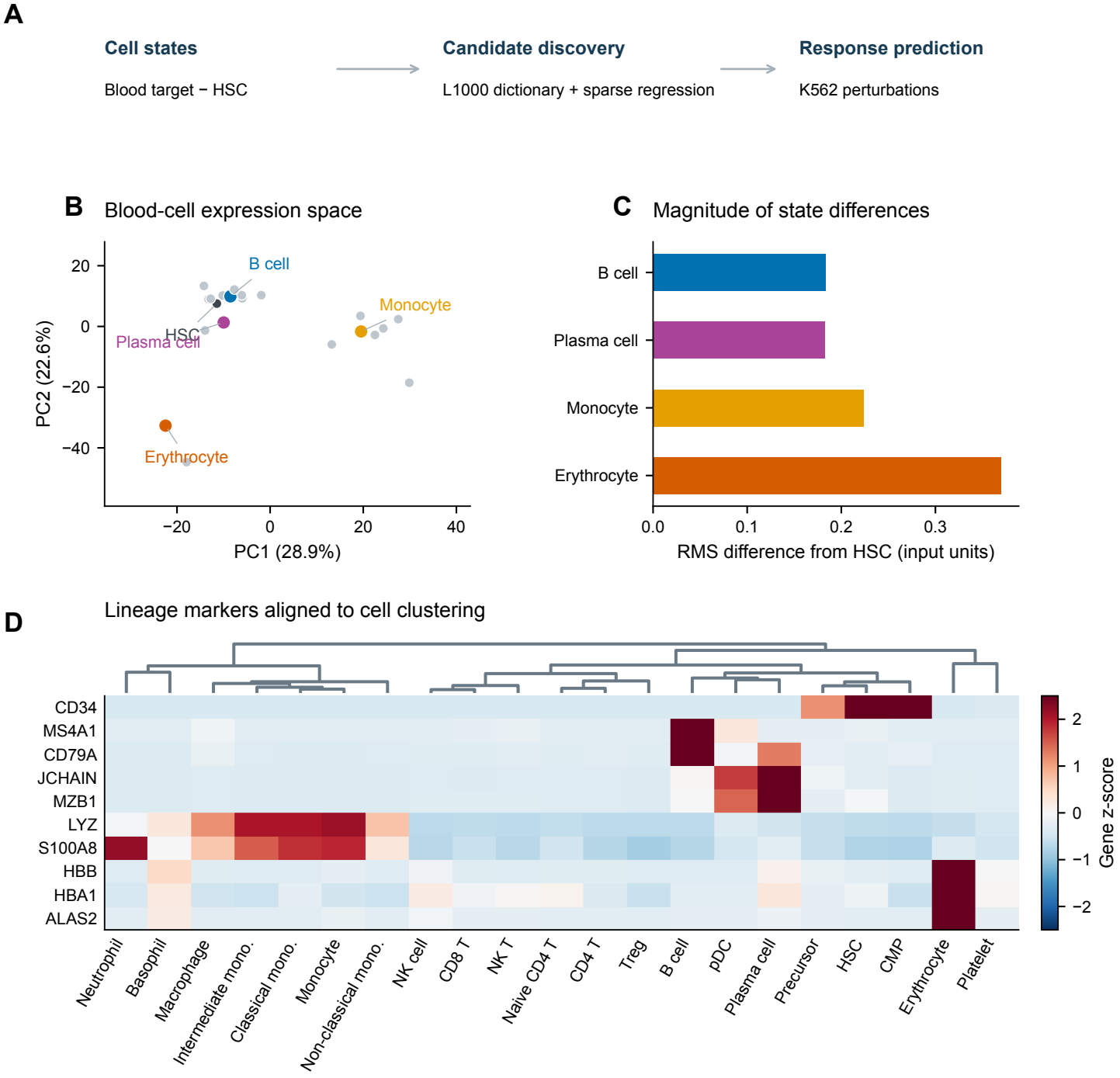

A
Cell states
Blood target − HSC
Candidate discovery
L1000 dictionary + sparse regression
Response prediction
K562 perturbations
B
Blood-cell expression space
B cell
Monocyte
HSC
Plasma cell
Erythrocyte
PC2 (22.6%)
PC1 (28.9%)
C
Magnitude of state differences
B cell
Plasma cell
Monocyte
Erythrocyte
RMS difference from HSC (input units)
D
Lineage markers aligned to cell clustering
CD34
MS4A1
CD79A
JCHAIN
MZB1
LYZ
S100A8
HBB
HBA1
ALAS2
Neutrophil
Basophil
Macrophage
Intermediate mono.
Classical mono.
Monocyte
Non-classical mono.
NK cell
CD8 T
NK T
Naive CD4 T
CD4 T
Treg
B cell
pDC
Plasma cell
Precursor
HSC
CMP
Erythrocyte
Platelet
Gene z-score